

Hypothesis 1 - Effect of Emotional Condition on Working-Memory Performance

Confirmatory Hypothesis Report

Hypothesis

Positive, negative, and neutral emotional conditions differentially affect working-memory performance.

Method

A trial-level confirmatory analysis was conducted to evaluate whether working-memory performance differed across neutral, positive, and negative emotional conditions. The analytical sample included 23 participants.

Response accuracy was analyzed using a generalized estimating equation (GEE) model with a binomial distribution, logit link, exchangeable within-participant correlation structure, and robust standard errors. Emotional condition was entered as the primary predictor, with the neutral condition specified as the reference category. The accuracy model included 17,820 trial-level observations clustered within 23 participants.

Reaction time was analyzed among valid correct-response trials only. Because reaction-time data were positively skewed, reaction time was log-transformed prior to modeling. A linear mixed-effects model was fitted with emotional condition as a fixed effect and a participant-specific random intercept. The reaction-time analysis included 14,184 observations from 23 participants. Maximum-likelihood estimation was used.

Model coefficients from the binomial GEE were exponentiated and interpreted as odds ratios. Coefficients from the log reaction-time model were exponentiated and converted to percentage changes relative to the neutral condition. Statistical significance was evaluated using a two-sided alpha level of .05, and 95% confidence intervals are reported.

Results

Accuracy

Descriptively, accuracy was highly similar across emotional conditions. Mean trial-level accuracy was 83.8% in the negative condition, 85.0% in the neutral condition, and 85.1% in the positive condition.

The omnibus Wald test indicated that emotional condition was not significantly associated with response accuracy, $\chi^2(2) = 2.54$, $p = .281$.

Relative to the neutral condition, the negative condition was associated with a small, non-significant reduction in the odds of a correct response ($\beta = -0.084$, $SE = 0.064$, $z = -1.31$, $p = .189$), corresponding to an odds ratio of approximately 0.92 (95% CI [0.81, 1.04]).

The positive condition did not differ from the neutral condition ($\beta = 0.002$, $SE = 0.057$, $z = 0.03$, $p = .979$), with an odds ratio of approximately 1.00 (95% CI [0.90, 1.12]).

Taken together, these findings provide no evidence that emotional condition substantially altered the probability of producing a correct response.

Table 1. Descriptive accuracy by emotional condition.

Condition	Trials	Accuracy	SD
Negative	5940	83.8%	0.368

Condition	Trials	Accuracy	SD
Neutral	5940	85.0%	0.357
Positive	5940	85.1%	0.356

Table 2. GEE estimates for response accuracy.

Contrast	OR	CI low	CI high	p
Negative vs Neutral	0.92	0.81	1.04	= 0.189
Positive vs Neutral	1.00	0.90	1.12	= 0.979

Reaction Time

In contrast to accuracy, emotional condition was strongly associated with reaction time.

Relative to the neutral condition, the negative emotional condition was associated with a significantly lower log reaction time (beta = -0.265, SE = 0.015, $z = -17.29$, $p < .001$, 95% CI [-0.295, -0.235]). After back-transformation, this coefficient corresponds to an estimated 23.3% reduction in reaction time relative to the neutral condition (approximately 20.9% to 25.5% lower).

The positive emotional condition was also associated with a significantly lower log reaction time compared with the neutral condition (beta = -0.285, SE = 0.015, $z = -18.60$, $p < .001$, 95% CI [-0.316, -0.255]). This corresponds to an estimated 24.8% reduction in reaction time relative to neutral (approximately 22.5% to 27.1% lower).

The magnitude of the positive and negative coefficients was similar, suggesting that both emotionally induced conditions were associated with faster responses compared with the neutral baseline.

Table 3. Mixed-effects model estimates for log reaction time.

Contrast	Beta	RT change	CI low	CI high	p
Negative vs Neutral	-0.265	-23.3%	-25.5%	-20.9%	< .001
Positive vs Neutral	-0.285	-24.8%	-27.1%	-22.5%	< .001

Decision rules

H1 supported if there is clear evidence of differences between emotional conditions in accuracy, RT, or both.

H1 partially supported if differences arise in only one outcome or in only one specific contrast.

H1 not supported if there is no evidence of relevant differences between conditions.

Interpretation

Hypothesis 1 was partially supported.

The emotional manipulation did not produce statistically detectable differences in response accuracy. Accuracy remained approximately 84-85% across the three emotional conditions, and the global test of emotional condition was not significant.

However, emotional condition showed a pronounced association with reaction speed. Both positive and negative emotional conditions were associated with substantially faster responses than the neutral condition, with estimated reductions in reaction time of approximately 25%.

Importantly, the faster responses under emotional induction were not accompanied by a statistically significant loss of accuracy. Thus, the results suggest that emotional induction influenced the speed component of working-memory performance more strongly than the probability of a correct response.

Because both positive and negative conditions showed effects in the same direction relative to neutral, these results do not yet demonstrate a valence-specific advantage of positive emotion or a valence-specific impairment under negative emotion. Those directional predictions should be evaluated through the planned contrasts in Hypotheses 2 and 3.

Hypothesis Assessment

Conclusion: Hypothesis 1 was partially supported. Emotional condition did not significantly affect accuracy, but both positive and negative emotional conditions were associated with substantially faster reaction times relative to the neutral condition.